

Chapter 4: Neutral Geometry

Projects

1. cite: Hartshorne, R.

"Geometry of Euclid and Beyond"

University of California, Berkeley (2000)

(from Hartshorne) "Let F be any ordered field"

$$\phi = \sum_{i \geq n}^6 a_i \cdot t^i \quad a_n \neq 0, a_i \in F, n \in \mathbb{Z}, \phi > 0$$

The field (F) is Non-Archimedean because

$$\phi = \{1, t, a_2 t^2, \dots, a_n t^n\} \text{ e.g. unequal spaced field.}$$

2. (pg 132) Archimedes' Axiom

"If CD is any segment, A any point, and r any ray with vertex A , then for every point $B \neq A$ on r there is a number n such that when CD is laid off n times on r starting at A , a point E is reached such that $n \cdot CD \cong AE$ and either $B = E$ or B is between A and E "

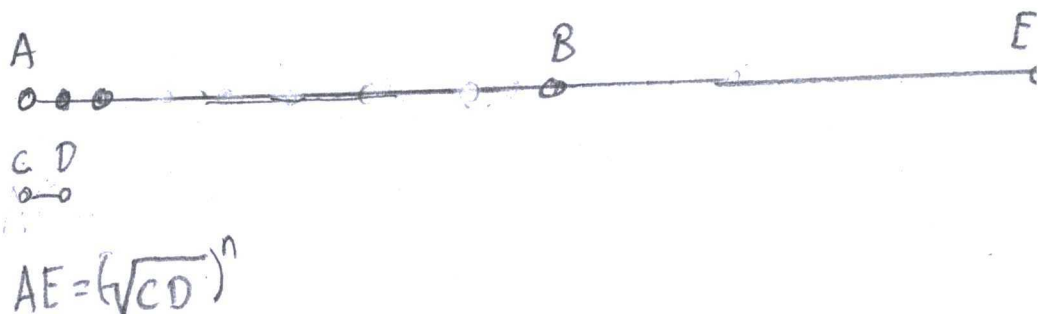
Euclidean Non-Archimedean field:

$$F = K(t_1) \cup K(t_2) \cup K(t_3) \cup \dots \cup K(t_n)$$

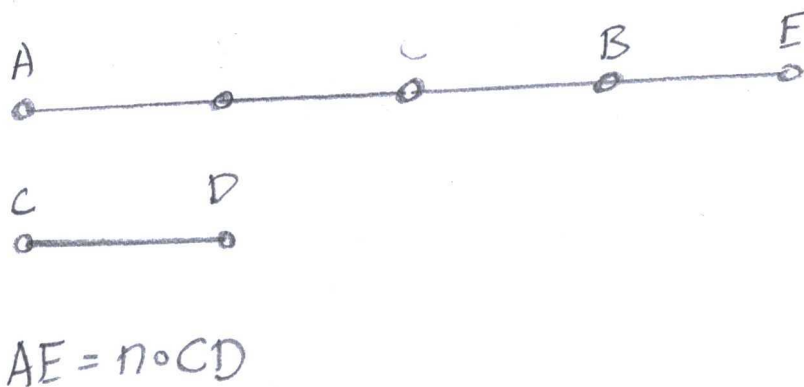
$$\text{where } t_1 = t \text{ and } t_{n+1} = \sqrt{t_n}$$

By image,

Euclidean Non-Archimedean Field:



Euclidean Archimedean Field:



3.

On page 196, Hartshorne described Euclid's area without numbers. Geometric shapes; circles, rectangles, squares; halves equalled after a line through center. Euclid's area related

lines by an interior point to a side and length. But I.35/36 comprehended badly because translation, specifically

Campanus(1220), University of Glasgow

Simsons(1834) then Heath(1997). Content

also borrowed from Arabic translations.

To end, half shapes overlapped without numbers.

4. Euclids proofs not discussed:

Book	Proof
I	4, 6, 7, 8, 12, 13, 14, 18, 19, 21, 25, 26, 28, 30, 31, 33, 35, 36, 37, 38, 43, 44-47
II	1, 2, 3, 5, 6, 7, 8
III	2-7, 8, 9, 11, 12, 13, 17, 19, 20, 29, 31
IV	All
V	1-3, 7, 8, 9, 10, 11-15, 18, 22

5. cyclic quadrilaterals are circumscribed by current definitions.

6. cite: Nelson, E.
"Syntax and Semantics"
Princeton, New Jersey (1973)

a) Edward commented on infinitesimals in history. The derivative contained infinitesimals by definition $\frac{f(x+h)-f(x)}{h}$. New interpretation changed formal notation. Calculus books removed infinitesimal notation toward a deductive syntax: $\lim_{h \rightarrow 0} \left| \frac{f(x+h)-f(x)}{h} \right| \cong d/dx$.

b) Edward begins a note about consistency, "Gödel established a completeness theorem."

Axioms are not consistent, and circular, such as the proof for commutation. necessitates commutation.